TCP Flow Control

(Receive Window, Speed Matching, and Buffer Management)

1. What is Flow Control?

TCP provides a **flow-control service** to eliminate the possibility of the sender overflowing the receiver's buffer. It is essentially a **speed-matching service** that balances the rate at which the sender transmits against the rate at which the receiving application reads.

Flow Control vs. Congestion Control

- **Flow Control:** Prevents a fast sender from overwhelming a slow **receiver**.
- **Congestion Control:** Prevents global network congestion by throttling the sender due to **network overload**.

2. The Receive Window (rwnd)

The sender maintains a variable called the **receive window** (`rwnd`), which indicates the current free space in the receiver's buffer.

Key Variables at the Receiver (Host B)

- `RcvBuffer`: Total size of the allocated receive buffer.
- `LastByteRead`: Number of the last byte read by the receiving application.
- `LastByteRcvd`: Number of the last byte arrived and placed in the buffer.

rwnd Calculation

$$\text{rwnd} = \text{RcvBuffer} - [\text{LastByteRcvd} - \text{LastByteRead}]$$

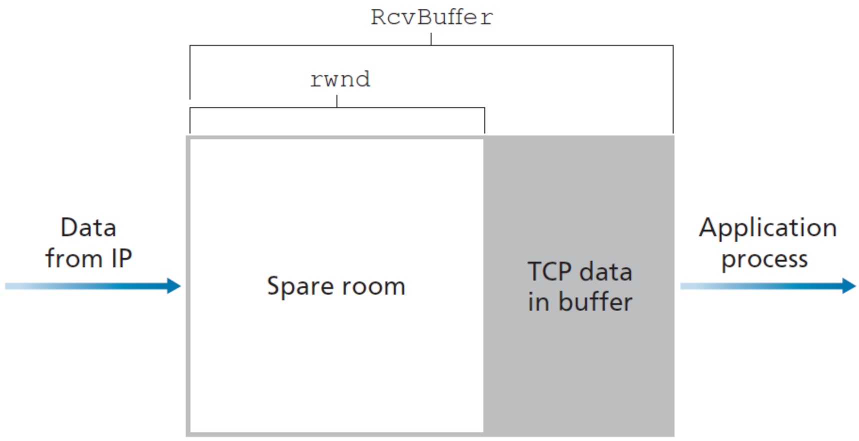

Figure 1: The Receive Window (`rwnd`) and Buffer Management

3. Sender Implementation

Host A (Sender) tracks two variables: `LastByteSent` and `LastByteAcked`. The amount of **unacknowledged data** must always be less than or equal to the receiver's advertised window.

Sender Constraint

$$\text{LastByteSent} - \text{LastByteAcked} \leq \text{rwnd}$$

3.1 The Zero Window Problem

If the receiver's buffer becomes completely full, it advertises `rwnd = 0`.

- If the receiver later clears space but has nothing to send (no data/ACKs), the sender remains blocked because it never receives an updated `rwnd` value.
- **The Solution:** TCP requires the sender to continue sending segments with **one data byte** even when `rwnd = 0`.
- These segments trigger ACKs from the receiver, eventually reclaiming a non-zero `rwnd` value when space opens up.

4. UDP Flow Control (The Lack Thereof)

Unlike TCP, **UDP does not provide flow control**.

- UDP segments are appended to a finite-sized queue preceding the socket.
- If the application does not read from the socket fast enough, the buffer overflows and segments are simply **dropped/lost**.

Network Designer Note

In UDP, data integrity and rate matching are the responsibility of the application layer.

5. Summary

Flow Control Highlights

- Speed matching between sender and receiving application.
- Managed via the dynamic `rwnd` field in TCP headers.
- Prevents buffer overflow at the destination end-system.
- Uses 1-byte probe segments to recover from zero-window states.